

ZERO TRUST ARCHITECTURE SIMULATION

Design and Simulation of a Zero Trust Architecture for Secure Enterprise Networking

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Main claim

A controlled Azure + Python lab can demonstrate core Zero Trust behavior: verify every request, enforce least privilege, and generate auditable monitoring evidence.

Perimeter trust breaks down when attackers get inside

Zero Trust changes the security question from “inside or outside?” to “should this exact request be allowed?”

Traditional model

Strong boundary controls access at entry
Internal network often receives broad trust
Compromised credentials can enable lateral movement
Flat network design increases blast radius

Risk created

One successful login or endpoint compromise may expose HR records, finance data, or administrative systems.



Zero Trust objective

Verify identity and account status
Check device compliance and risk
Enforce role and source-segment rules
Generate logs and alerts for every denial

13

policy test cases

3

allowed requests

10

denied requests

Design goal: show core Zero Trust behavior using real cloud configuration evidence and reproducible simulation output.

The access decision combines identity, device posture, segment, and monitoring

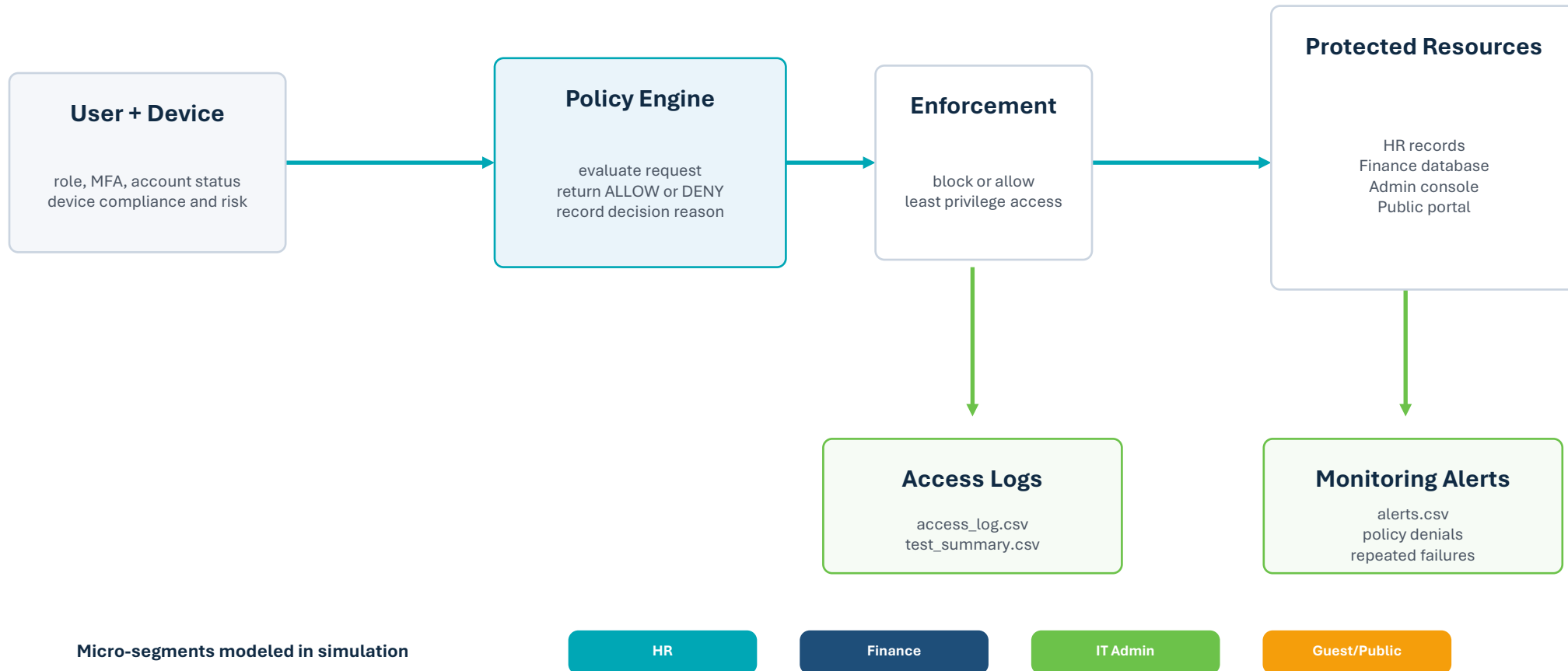


Core rule implemented in Python

ALLOW only when no policy condition fails; otherwise DENY and record the reason.

Least privilege: role/resource match • Micro-segmentation: allowed source path • Monitoring: denials generate alerts

The prototype separates enforcement, policy decisions, resources, and monitoring

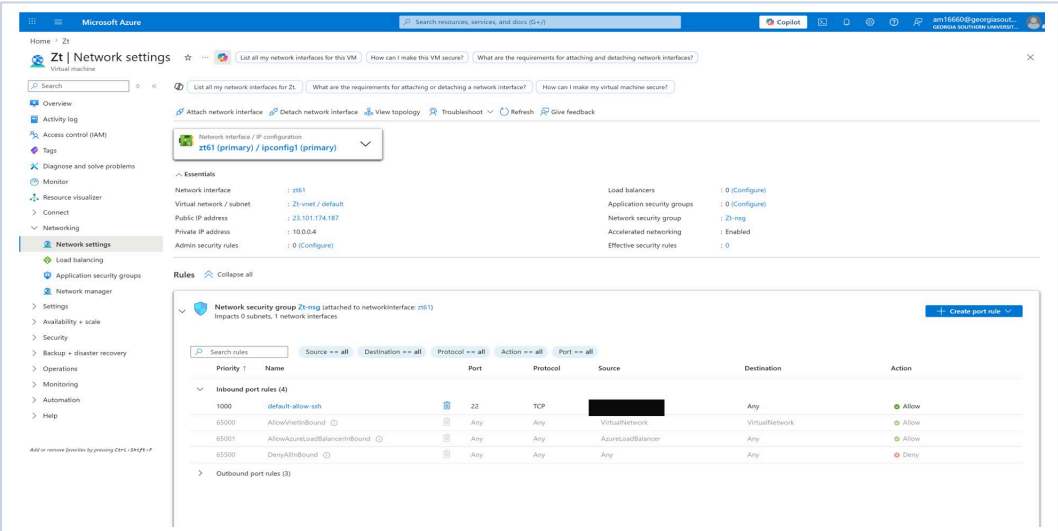


Architecture is a small-scale prototype: policy logic + cloud lab evidence, not a production Zero Trust platform.

The Azure VM provided a real Linux environment for secure testing

Deployment details

Resource	Type	Status
Zt	Microsoft.Compute/virtualMach	OK
zt61	Microsoft.Network/networkInter	OK
Zt-nsg	Network security group	OK
Zt-vnet	Virtual network	OK
Zt-ip	Public IP address	OK



1

Cloud resources created

VM, NIC, NSG, VNet, and public IP were created as a controlled Azure lab.

2

SSH restricted

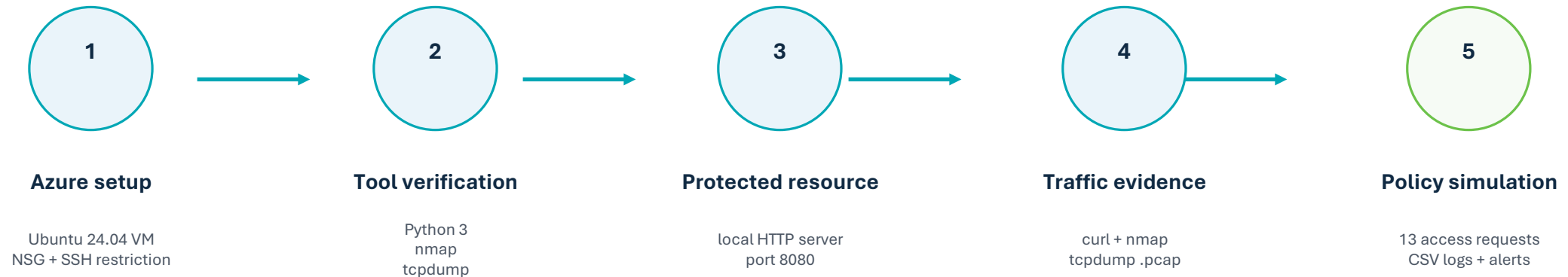
Port 22 stayed available for administration, but the source was limited to a trusted /32 public IP.

3

Security purpose

The lab demonstrates secure remote access before moving into packet capture and simulation.

Implementation connected cloud setup, traffic generation, and policy simulation



Selected command evidence

```
python3 src/zero_trust_simulation.py → access_log.csv + alerts.csv + test_summary.csv
```

The workflow keeps testing controlled: all scans and packet captures target the VM/localhost resource, not external systems.

Packet capture confirmed real HTTP traffic was generated and inspected

```

azureuser@Zt:~/zt_lab/www$ sudo tcpdump -nn -r ~/zt_lab/evidence/zt_capture.pcap | head -20
reading from file /home/azureuser/zt_lab/evidence/zt_capture.pcap, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
Warning: interface names might be incorrect
10:48:14.168227 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [S], seq 1320977220, win 65495, options [mss 65495,sackOK,TS val 2197598675 ecr 0,nop,wscale 7], length 0
10:48:14.168237 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [S.], seq 2833041329, ack 1320977221, win 65483, options [mss 65495,sackOK,TS val 2197598675 ecr 2197598675,nop,wscale 7], length 0
10:48:14.168248 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 1, win 512, options [nop,nop,TS val 2197598675 ecr 2197598675], length 0
10:48:14.168307 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [P.], seq 1:78, ack 1, win 512, options [nop,nop,TS val 2197598675 ecr 2197598675], length 77: HTTP: GET / HTTP/1.1
10:48:14.168314 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [.], ack 78, win 511, options [nop,nop,TS val 2197598675 ecr 2197598675], length 0
10:48:14.168862 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [P.], seq 1:186, ack 78, win 512, options [nop,nop,TS val 2197598676 ecr 2197598675], length 185: HTTP: HTTP/1.0 200 OK
10:48:14.168880 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 186, win 511, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168912 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [P.], seq 186:233, ack 78, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 47: HTTP
10:48:14.168918 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 233, win 511, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168945 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [F.], seq 78, ack 233, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168953 lo In IP 127.0.0.1.8080 > 127.0.0.1.34218: Flags [F.], seq 233, ack 79, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0
10:48:14.168967 lo In IP 127.0.0.1.34218 > 127.0.0.1.8080: Flags [.], ack 234, win 512, options [nop,nop,TS val 2197598676 ecr 2197598676], length 0

```

WHAT THE CAPTURE SHOWS

TCP handshake to port 8080
 HTTP GET request to local resource
 HTTP/1.0 200 OK response
 Saved .pcap file read back with tcpdump

Why it matters

This validates the monitoring layer with real packet evidence, not only simulated logs.

Traffic was generated only against the local protected resource on 127.0.0.1:8080.

The simulation denies access unless every policy condition passes

Simplified policy logic

```
if user.role not in resource.allowed_roles:  
    reasons.append("role not authorized")  
  
if resource.requires_managed_device and not  
device.managed:  
    reasons.append("managed device required")  
  
if request.source_segment not in allowed_sources:  
    reasons.append("blocked by segmentation")  
  
decision = "ALLOW" if not reasons else "DENY"
```

Decision = ALLOW only when the reasons list stays empty.

1

Identity

User exists, account is active, role is allowed, MFA requirement is satisfied.

2

Device posture

Managed, compliant, encrypted when sensitive, and below risk threshold.

3

Micro-segmentation

The source segment must be authorized for the requested resource.

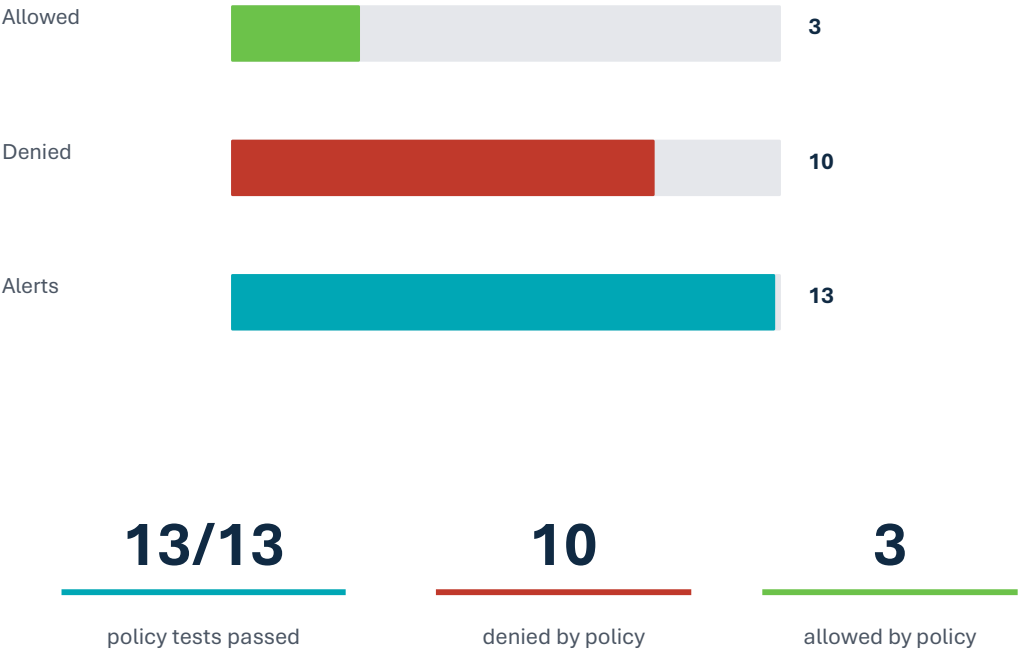
4

Monitoring

Denied requests generate alerts; repeated denials escalate severity.

All 13 policy tests passed, with risky requests denied by design

Result counts



```
azureuser@Zt:~/zt_lab$ cat results/test_summary.csv
request_id,scenario,expected_decision,actual_decision,policy_result
T01,HR user accesses HR records from compliant device,ALLOW,ALLOW,PASS
T02,HR user attempts Finance database access,DENY,DENY,PASS
T03,Finance user uses noncompliant device,DENY,DENY,PASS
T04,Admin accesses admin console from trusted segment,ALLOW,ALLOW,PASS
T05,Guest attempts internal HR access,DENY,DENY,PASS
T06,High-risk unmanaged device attempts admin access,DENY,DENY,PASS
T07,Valid HR user attempts access from wrong segment,DENY,DENY,PASS
T08A,Repeated failed attempt 1,DENY,DENY,PASS
T08B,Repeated failed attempt 2,DENY,DENY,PASS
T08C,Repeated failed attempt 3,DENY,DENY,PASS
T09,Device remains noncompliant after prior denial,DENY,DENY,PASS
T10,Guest accesses public portal,ALLOW,ALLOW,PASS
T11,Inactive account attempts access,DENY,DENY,PASS
```

Test categories included cross-segment access, noncompliant devices, high-risk devices, public access, and inactive accounts.

The monitoring engine turns denied access into auditable alerts

```

azureuser@Zt:~/zt_lab$ cat results/alerts.csv
timestamp,request_id,severity,alert_type,message
2026-05-09T22:38:40.734972+00:00,T02,MEDIUM,POLICY_DENIAL,Access denied for u_hr to FINANCE_DB: User
role is not authorized for this resource; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.734981+00:00,T03,MEDIUM,POLICY_DENIAL,Access denied for u_fin to FINANCE_DB: Dev
ice is not compliant
2026-05-09T22:38:40.734993+00:00,T05,MEDIUM,POLICY_DENIAL,Access denied for u_guest to HR_RECORDS: U
ser role is not authorized for this resource; MFA is required but not enabled; Managed device is req
uired; Encrypted device is required for sensitive resource; Source segment is blocked by micro-segme
ntation policy
2026-05-09T22:38:40.734998+00:00,T06,MEDIUM,POLICY_DENIAL,Access denied for u_guest to ADMIN_CONSOLE
: User role is not authorized for this resource; MFA is required but not enabled; Managed device is
required; Device is not compliant; Encrypted device is required for sensitive resource; Device risk
score is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735003+00:00,T07,MEDIUM,POLICY_DENIAL,Access denied for u_hr to HR_RECORDS: Sour
ce segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735006+00:00,T08A,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB:
User role is not authorized for this resource; MFA is required but not enabled; Managed device is re
quired; Device is not compliant; Encrypted device is required for sensitive resource; Device risk sc
ore is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735011+00:00,T08B,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB:
User role is not authorized for this resource; MFA is required but not enabled; Managed device is re
quired; Device is not compliant; Encrypted device is required for sensitive resource; Device risk sc
ore is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735011+00:00,T08B,HIGH,REPEATED_DENIALS,Device d_unknown triggered repeated deni
ed access attempts
2026-05-09T22:38:40.735017+00:00,T08C,MEDIUM,POLICY_DENIAL,Access denied for u_guest to FINANCE_DB:
User role is not authorized for this resource; MFA is required but not enabled; Managed device is re
quired; Device is not compliant; Encrypted device is required for sensitive resource; Device risk sc
ore is too high; Source segment is blocked by micro-segmentation policy
2026-05-09T22:38:40.735017+00:00,T08C,HIGH,REPEATED_DENIALS,Device d_unknown triggered repeated deni
ed access attempts
2026-05-09T22:38:40.735022+00:00,T09,MEDIUM,POLICY_DENIAL,Access denied for u_fin to FINANCE_DB: Dev
ice is not compliant
2026-05-09T22:38:40.735028+00:00,T11,MEDIUM,POLICY_DENIAL,Access denied for u_disabled to HR_RECORDS
: Inactive user account; Device is not assigned to requesting user
2026-05-09T22:38:40.735028+00:00,T11,HIGH,REPEATED_DENIALS,Device d_hr_laptop triggered repeated deni
ed access attempts

```

SIMULATED RESPONSE WORKFLOW



Alert types produced

POLICY_DENIAL + REPEATED_DENIALS

The prototype met the project objective, with clear paths to scale

What worked

Azure lab gave real Linux evidence
SSH restriction reduced exposure
Packet capture validated monitoring
Simulation produced 13/13 passing tests

Limitations

Single VM prototype
No real identity provider
No production SIEM integration
No Snort/Suricata IDS deployment

Future scale-up

Multi-VM segmented lab
Microsoft Entra / Okta integration
SIEM dashboard and alert correlation
IDS-based detection with Snort or Suricata

Bottom line: the project demonstrated core Zero Trust behavior in a controlled, reproducible environment, while leaving room to expand into a larger enterprise-style security lab.

Final takeaway

Zero Trust is strongest when policy, evidence, and monitoring work together.

Submission artifacts

● Final report

● Screenshot evidence packet

● Source code zip

● Live demo recording
uploaded with project requirements

● Presentation deck

Recorded demo shows the simulation run, result files, policy summary, alerts, and access log.

Q&A

Thank you